

CSE273: Foundations of Machine Learning

Unit-1, Part-1: Machine Learning Foundation (Concepts and Context of ML Landscape)



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The Story of a Spam Filter



The Story of a Spam Filter

- Imagine you've been hired by a major email provider to block "spam" emails.
- Initially, you write several of rules, like:
 - "If the subject contains 'WINNER', it's spam"
 - "If the sender is unknown and asks for money, it's spam"
 - ...

- **But the spammers are clever.**

- They start writing "W-I-N-N-E-R" or "Congratulations!"
- 📧 Your rules fail. You're exhausted.
- Then, You realize **you can't manually code every possibility.**



The Story of a Spam Filter

- **You realize you can't manually code every possibility.**
 - Instead, you show the computer 10,000 examples of real spam and 10,000 real emails.
 - The **computer notices (or, learns)** that spam emails often have certain patterns of words you hadn't even thought of.



- Suddenly, the system starts blocking spam it has never seen before. That is the **heart of Machine Learning.** 💖



How AI Beats Spam: From Manual Rules to Machine Learning

The Old Way: Manual Rules



The New Way: Machine Learning

Feed the computer thousands of examples.
Show it 10,000 spam emails and 10,000 legitimate emails.



The computer learns hidden patterns on its own.
It spots connections humans might not have thought of.



The system can now block new, unseen spam.
This is the core power of Machine Learning.

The Story of a Spam Filter

- **Traditional Programming:**
 - You write explicit "if-then" rules.
 - If the rules are wrong, the system fails.
- **Machine Learning:**
 - We provide the system with **Experience (E)** (past emails) for a **Task (T)** (identifying spam) to improve its **Performance (P)** (accuracy).

Traditional

- Explicit "if-then" rules
- The rules go wrong
→ System Fails

Machine Learning

- Experience (E)
- Task (T)
- Performance (P)

Introduction to Machine Learning

Defining Machine Learning

1959

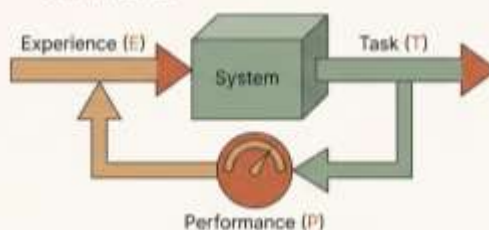
“Field of study that gives computers the ability to learn without being explicitly programmed.”

— Arthur Samuel
Inter

1997

A computer program is said to learn from Experience (E) with respect to some class of Tasks (T) and Performance measure (P), if its performance at tasks in T, as measured by P, improves with experience E.

— Tom Mitchell

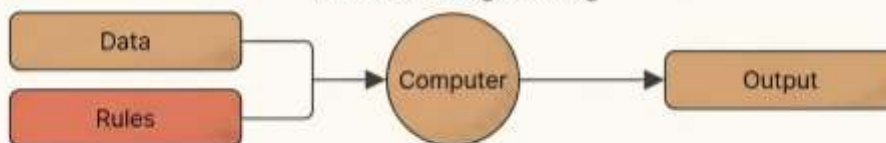


Introduction to Machine Learning

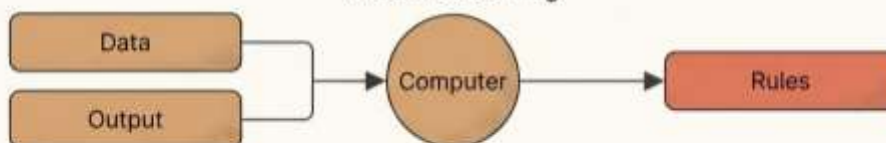
The Paradigm Shift

From Logic to Learning

Traditional Programming

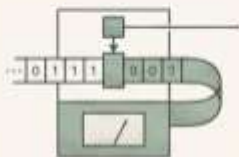


Machine Learning



In traditional programming, humans hard-code the rules. In Machine Learning, the system infers the rules by analyzing the relationship between data and answers.

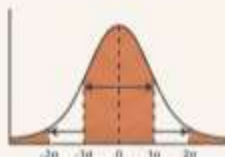
A Brief History of Intelligence



1950s - The Dawn

Turing Test & Perceptrons

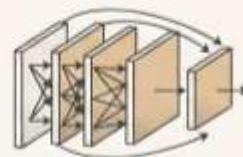
The fundamental question:
"Can machines think?"



1980s - The Probabilistic Turn

Rise of Statistical Learning

Moving from rigid logic to probability-based models.



2010s - The Deep Learning Boom

Big Data + GPUs

Massive datasets (ImageNet) enable deep neural networks.

THE TURING TEST (IMITATION GAME)

Evaluating AI's ability to exhibit human-like intelligence through text-based conversation.

THE JUDGE (HUMAN)



Asks questions
via text interface.

HUMAN PARTICIPANT



Answers honestly to
convince judge.

MACHINE (AI)

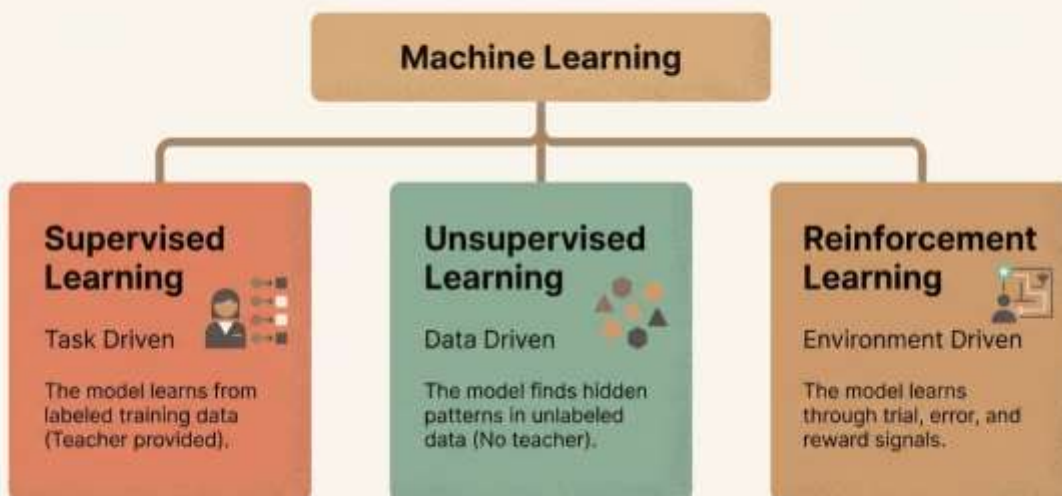


Tries to fool the judge by
imitating human responses.

Goal: If the Judge cannot reliably tell the machine
from the human, the machine is said to pass.

Types of Machine Learning

The Taxonomy of Learning



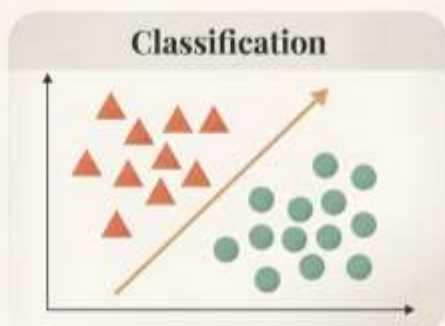
Note: Semi-supervised learning exists as a hybrid approach.

Types of Machine Learning

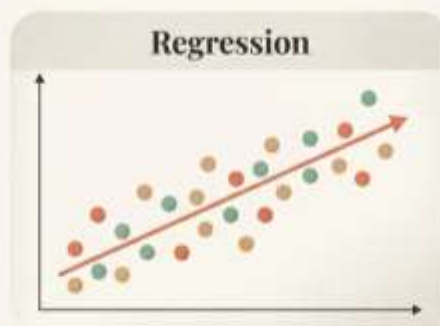
Supervised Learning

Learning with a Teacher

The algorithm is trained on a labeled dataset consisting of inputs (X) and correct outputs (Y). The goal is to learn a mapping function to predict Y for new X.



Predicting a Category
(e.g., Spam vs. Not Spam)



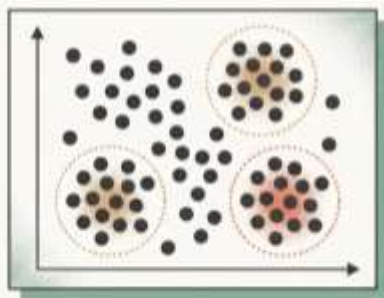
Predicting a Continuous Value
(e.g., House Prices)

Types of Machine Learning

Unsupervised Learning

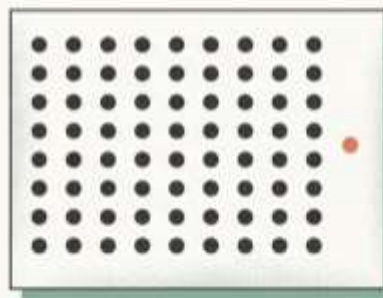
Finding Hidden Structures

The algorithm is given data without explicit instructions or labels. It must explore the data to find inherent structures.



Clustering

Grouping similar data points (e.g., Customer Segmentation)



Anomaly Detection

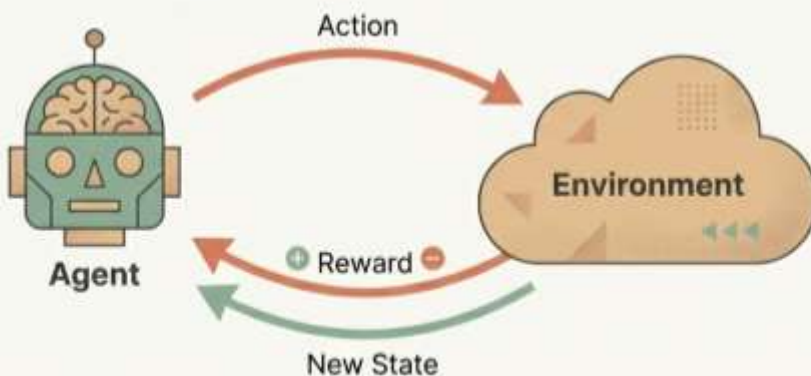
Identifying outliers (e.g., Fraud Detection)

Types of Machine Learning

Reinforcement Learning

Interaction and Adaptation

An agent learns to make decisions by performing actions in an environment and receiving feedback.



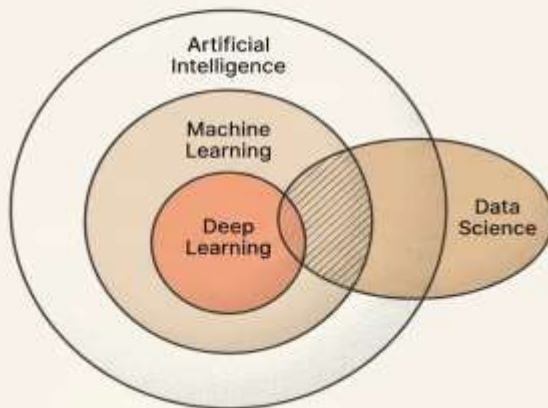
Examples:

- Game Playing (Chess, Go)
- Robotics (Navigation)
- Self-driving Cars (Trajectory Optimization)

AI vs ML vs Data Science

Mapping the Territory

Distinguishing AI, Machine Learning, and Data Science



Artificial Intelligence

Any technique creating machines that mimic human behavior and intelligence.

Machine Learning

A subset of AI: The practice of using algorithms to parse data, learn from it, and make a determination.

Data Science

An interdisciplinary field combining Domain Expertise, Programming, and Math to extract insights from data.

AI vs ML vs Data Science

The story of a Smart Hospital

- Imagine you are building a futuristic "**Smart Hospital**".

The AI Vision

- You want a robotic assistant that can walk through halls, talk to patients, and decide which emergency needs attention first.
- This robot is **mimicking human-like behavior and reasoning**.

The ML Engine

- To make this robot useful, it needs to recognize diseases in X-rays.
- You don't program "**if pixel is red, it's a tumor**".
- Instead, you feed it 50,000 X-rays so it can **learn the patterns of a tumor on its own**.

The Data Science Insights

- Meanwhile, the hospital administrator asks, "**Why are patients waiting so long on Tuesdays?**"
- You dive into the logs, **clean up the messy data, create visualizations, and discover** that a specific lab is understaffed during lunch hours. You've just **extracted a "business insight"** from data.

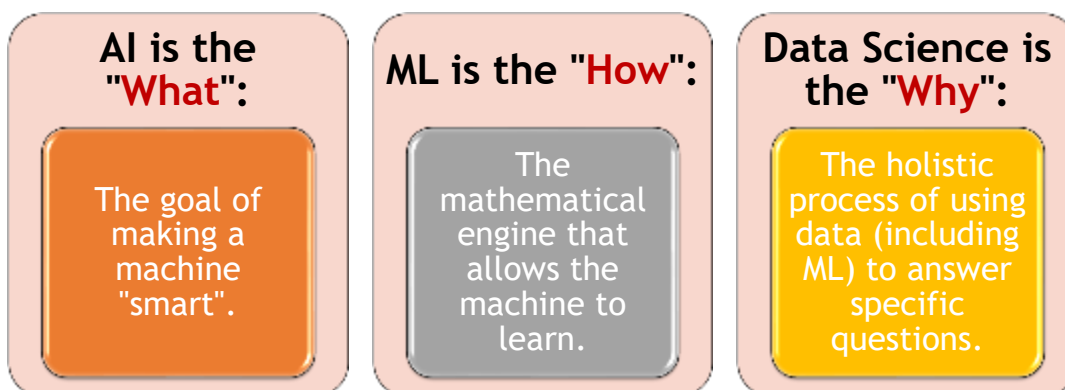
AI vs ML vs Data Science

While these fields overlap significantly, their core objectives and methodologies differ. Here is a breakdown...

Feature	Artificial Intelligence (AI)	Machine Learning (ML)	Data Science (DS)
Core Definition	Systems that mimic human intelligence to perform tasks.	A subset of AI that learns patterns from data without explicit programming.	An interdisciplinary field used to extract insights and "meaning" from data.
Primary Goal	To create "smart" machines that can reason and act.	To build models that improve their performance as they see more data.	To find hidden patterns and actionable insights to solve problems.
Data Type	Often uses rules and logic; doesn't always need "big data".	Highly dependent on large, high-quality structured/unstructured data.	Handles the entire data pipeline (cleaning, processing, and analyzing).
Key Tools	Logic engines, Neural Networks, Robotics.	NumPy, Scikit-Learn, TensorFlow, PyTorch.	Pandas, Matplotlib, SQL, Tableau.
Real-World Example	A humanoid robot or a self-driving car's decision-making system.	A Netflix "Recommended for You" list or a Spam Filter.	Analyzing 5 years of sales data to predict next year's budget.

AI vs ML vs Data Science

Key Takeaways!



- In this course, we will spend a lot of time on **Unit-1**, where we focus on the ML part of this trio, then we will use the tools of Data Science (**Pandas and Matplotlib**) in **Unit-2** to make sure our models are actually seeing the right things in the data.

Revisiting The Paradigm Shift: From Manual Rules to Learning

The Manual Approach (Rules)



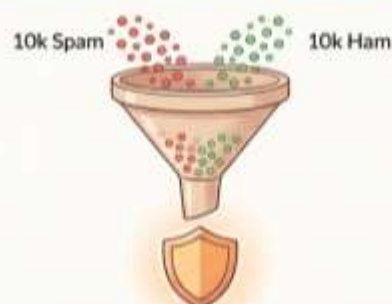
Context: A developer writes thousands of explicit rules.

Example: If subject == 'WINNER': block()

The Failure: Spammers adapt (e.g., 'W-I-N-N-E-R').

Manual coding cannot handle infinite variations.

The Machine Learning Approach (Data)



Action: Injecting data instead of rules.

Mechanism: The system identifies hidden patterns (frequencies, word combos) invisible to humans.

Result: Generalization. The model handles unseen variations.

Core Insight: Traditional programming feeds rules to process data. Machine Learning feeds data to derive rules.

Defining Structure: Parametric vs. Non-Parametric Models

Parametric Models



"Fitting a line."

- **Definition:** Assumes a fixed functional form (shape) for the data. The number of parameters is fixed, regardless of data size.
- **Pros:** Faster training, interpretable.
- **Cons:** Limited flexibility (Risk of High Bias).
- **Examples:** Linear Regression, Logistic Regression.

Non-Parametric Models



"Connect-the-dots."

- **Definition:** Does not assume a fixed shape. The model structure grows and complexity increases as data volume increases.
- **Pros:** Highly flexible, accurate for complex patterns.
- **Cons:** Slower, requires more data (Risk of Overfitting).
- **Examples:** K-Nearest Neighbors (KNN), Decision Trees.

Machine Learning Models

(Based on Goal)

Learning Goals: Generative vs. Discriminative Models

Generative Models



 **The Math:** Models Joint Probability $P(X, Y)$

 **Goal:** Learn the underlying distribution of each class to understand how data is generated.

 **Analogy:** Learning to draw a cat from memory.

 **Example:** Naive Bayes, Hidden Markov Models.

Discriminative Models



 **The Math:** Models Conditional Probability $P(Y | X)$

 **Goal:** Learn the boundary or decision surface that separates the classes.

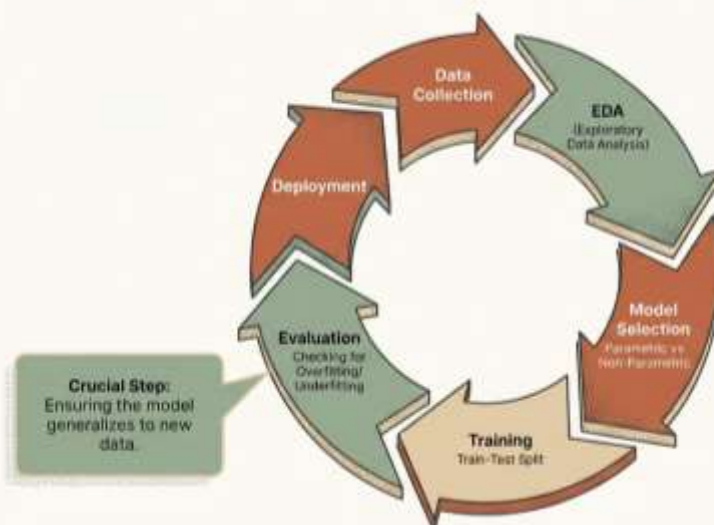
 **Analogy:** Learning to distinguish a cat from a dog.

 **Example:** Logistic Regression, SVM.

Machine Learning Workflow

Based on Components of a Learning System

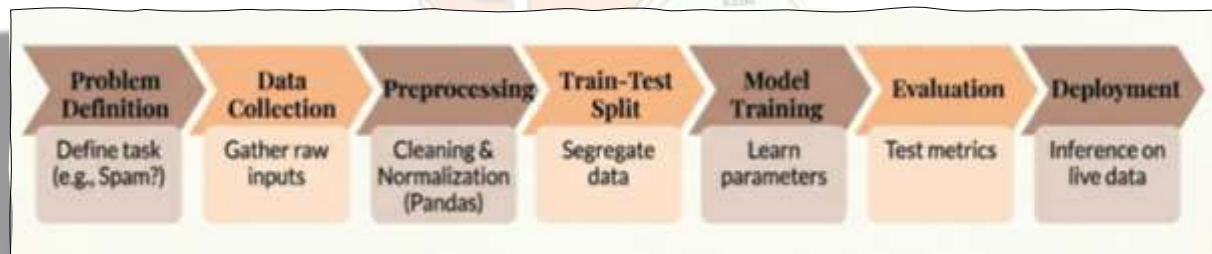
From Raw Data to Deployment



Machine Learning Workflow

Based on Components of a Learning System

From Raw Data to Deployment



The Train-Test Data Split

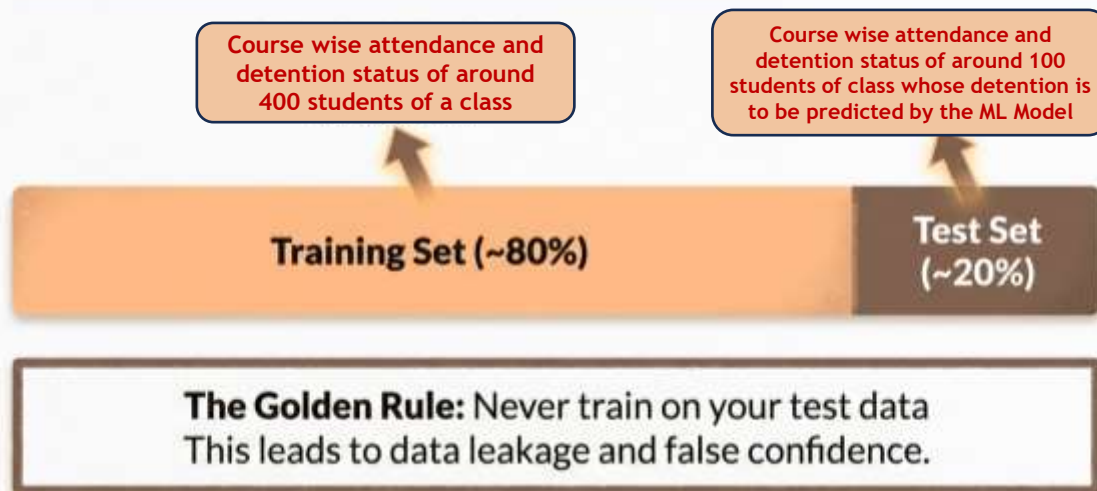
Validating Generalization



The Golden Rule: Never train on your test data
This leads to data leakage and false confidence.

The Train-Test Data Split

Validating Generalization



ML Model Evaluation Metrics



Classification Metrics (Categorical Output)

Accuracy: Overall % correct. (Misleading if classes are imbalanced)

Precision: Quality: Of all labeled spam, how many were actually spam?

Recall: Quantity: Of all actual spam, how many did we catch?

F1-Score: Harmonic mean of Precision and Recall.



Regression Metrics (Continuous Output)

Mean Squared Error (MSE): Average of squared differences.
Penalizes large errors heavily.

Mean Absolute Error (MAE): Average of absolute differences.
More robust to outliers.

Evaluation Metrics for Classification Models

Beyond Accuracy: Measuring the performance of the Spam Detective



The Old Way: Rules



If subject = 'WINNER' → Spam.
If unknown sender → Spam.
Result: Spammers adapted, rules failed.

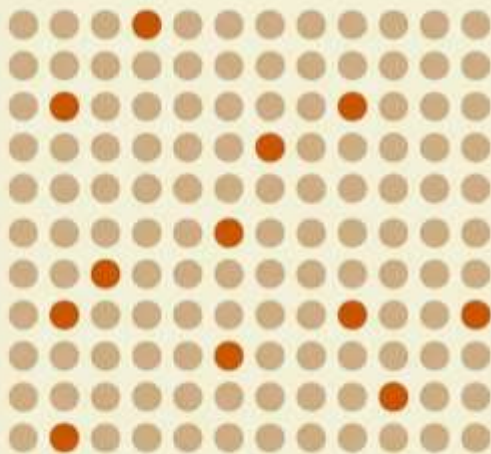
The New Way: Machine Learning



Input: 10,000 real examples.
Result: The computer found hidden patterns and blocked new spam variants.

The computer found the patterns. Now, we must grade how well it works.

The Scenario: 100 Incoming Emails



Total Emails: 100

Actual Safe (Ham): 90 

Actual Spam: 10 

Note: This is an imbalanced dataset. Real spam is rare compared to normal mail.

The Anatomy of the Confusion Matrix

		PREDICTED CLASS	
		Spam	Safe
ACTUAL CLASS	Spam	True Positive (TP) The Thief Caught	False Negative (FN) The Thief Escaped
	Safe	False Positive (FP) The Innocent Arrested	True Negative (TN) The Citizen Left Alone

The Model's Performance Data

		PREDICTED CLASS	
		Spam	Safe
ACTUAL CLASS	Spam	9 Caught Spam	1 Missed Spam
	Safe	9 False Alarm	81 Correctly Ignored

Total Predicted Spam: 18 | Total Actual Spam: 10

ML Model Evaluation Metrics

	PREDICTED CLASS	
	Spam	Safe
ACTUAL CLASS	Spam	9 Caught Spam
	Safe	1 Missed Spam
	Spam	9 False Alarm
	Safe	81 Correctly Ignored

Total Predicted Spam: 18 | Total Actual Spam: 10

The **Accuracy** Trap

90%

Formula: $(\text{True Positives} + \text{True Negatives}) / \text{Total}$

Calculation: $(9 + 81) / 100 = 0.90$

The Trap:

90% sounds excellent. But imagine a "lazy" model that simply marks EVERY email as Safe. Because 90% of our mail is safe, that lazy model would also score 90% accuracy.

Takeaway: Accuracy is deceptive on imbalanced datasets.

ML Model Evaluation Metrics

Precision: Can we trust the alarm?

	PREDICTED CLASS	
	Spam	Safe
ACTUAL CLASS	Spam	9 Caught Spam
	Safe	1 Missed Spam
	Spam	9 False Alarm
	Safe	81 Correctly Ignored

Definition: When the model screams 'SPAM', how often is it right?

Formula: $TP / (TP + FP)$

Calculation: $9 / (9 + 9) = 50\%$

Narrative:

Our model flagged 18 emails as spam, but was only right half the time. This means 50% of the alerts were false alarms, annoying the user.



ML Model Evaluation Metrics

Recall: Did we catch them all?

		Predicted Class	
		Spam	Safe
Actual Class	Spam	Caught Spam 9 TP	Missed Spam 1 FN
	Safe	9 False Alarm	81 Correctly Ignored

Definition: Out of all the actual spam, how much did we find?

Formula: $TP / (TP + FN)$

Calculation: $9 / (9 + 1) = 90\%$

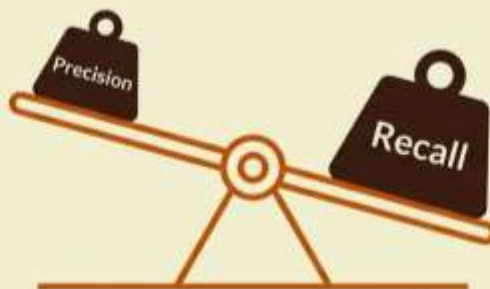
Narrative: We found 90% of the spam. The model acts as an effective dragnet, letting very few threats slip through.



ML Model Evaluation Metrics

The Precision-Recall Trade-off

To increase **Precision**
(avoid **false alarms**),
we must be
conservative.



To increase **Recall**
(**catch all spam**), we
must be aggressive.

Our Current Model: Aggressive. High Recall (90%) but Low Precision (50%).

F1 Score: The Harmonic Balance

The **F1 Score** is the **harmonic mean** of **Precision** and **Recall**. It penalizes extreme values, providing a single score that represents the balance.

$$F1 = 2 * \frac{(\text{Precision} * \text{Recall})}{\text{Precision} + \text{Recall}}$$

$$2 * (0.5 * 0.9) / (0.5 + 0.9) = \mathbf{0.64}$$

Verdict: While Accuracy was 90%, the F1 Score of 0.64 reveals the model is actually mediocre because of the poor Precision.

Specificity: The True Negative Rate

	Predicted Spam	Predicted Safe
Actual Spam	91	9
Actual Safe	9 FP	81 TN

Definition: How effective is the model at leaving safe emails alone?

Formula: $TN / (TN + FP)$

Calculation: $81 / (81 + 9) = \mathbf{90\%}$

Context: 90% Specificity means 10% of safe emails are being wrongly flagged. In a business context, losing 1 in 10 legitimate emails is a major problem.

ML Model Evaluation Metrics

Model Performance Report

	Predicted Spam	Predicted Safe
Actual Spam	9	9
Actual Safe	1 FP	81 TN

Accuracy: 90% (Misleading)



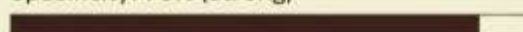
Precision: 50% (Poor)



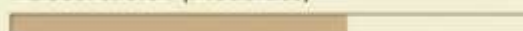
Recall: 90% (Strong)



Specificity: 90% (Strong)



F1 Score: 0.64 (Moderate)



This model is safe for a 'Junk' folder, but too aggressive to delete emails automatically.

ML Model Evaluation Metrics

Choosing the Right Metric



Scenario A: Spam Filter

Cost of False Positive is High
(Lost Mail)

Goal: High Precision

We accept some spam to protect
real mail.



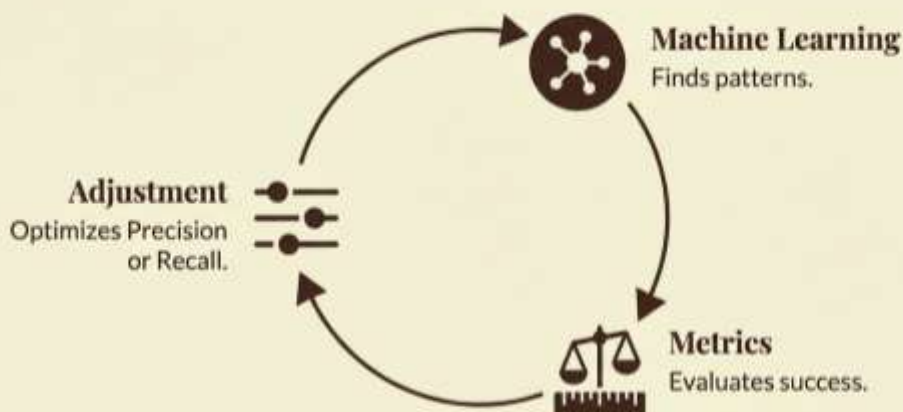
Scenario B: Disease Detection

Cost of False Negative is High
(Missed Diagnosis)

Goal: High Recall

We accept false alarms to ensure we
don't miss a patient.

The Loop of Improvement



A model with 99% accuracy can be useless if it fails the specific metric that matters to your users. Always look beneath the top-line number.

Reference: Metric Cheat Sheet

Accuracy: $(TP+TN) / \text{Total}$	→ "Trustworthiness"
Precision: $TP / (TP+FP)$	→ "Coverage"
Recall (Sensitivity): $TP / (TP+FN)$	→ "True Negative Rate"
Specificity: $TN / (TN+FP)$	→ "True Negative Rate"
F1 Score: $2 * (\text{Prec} * \text{Rec}) / (\text{Prec} + \text{Rec})$	→ "Harmonic Mean"



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Try
Yourself!

ML Model Evaluation Metrics

NAAC
GRADE A++

Is Your AI a Good Sentry? A Guide to Machine Learning Metrics

The Sentry & The Wolf: An AI Analogy

Correct Calls



True Positive
(Threat Detected)

Critical Mistakes



False Positive
(A "False Alarm")



True Negative
(Peaceful Night)



False Negative
(A "Missed Danger")

How We Grade the Sentry (Key Metrics)



Precision

How often was the alarm real?
Prioritize this when the cost of a false alarm is high.



Recall

Did we catch every wolf?
Prioritize this when missing a single threat is disastrous.



Accuracy

What was the overall success rate?
Can be misleading if wolves (threats) are very rare.

Professor's Challenge: Credit Card Fraud



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ML Model Evaluation Metrics

NAAC
GRADE A++

Evaluation Metrics for Regression Models

- When we are predicting a continuous number (like house prices), we measure how far off our arrows (**predictions**) are from the bullseye (**expected result**).

Metric	Description	Key Characteristics	Error Sensitivity
MAE (Mean Absolute Error)	Average of the absolute errors.	Treat all errors equally; robust to outliers.	Low; it does not disproportionately penalize large errors.
MSE (Mean Squared Error)	Average of the squared errors.	Penalizes large errors heavily; very useful for optimization.	High; large errors are squared, making them much more significant.
RMSE (Root MSE)	The square root of MSE.	Returns the error value to the same units as our data (e.g., "Dollars").	High; like MSE, it is sensitive to large errors due to the squaring process.
R² Score	The proportion of variance explained.	Tells you how much better your model is than just guessing the average.	N/A; measures fit quality relative to a baseline mean model.

The Archer's Ledger

Measuring Precision in Regression Models



A visual guide to MAE, MSE, RMSE, and R^2 using real estate prediction.

Based on "The Sentry's Guide to Machine Learning Evaluation Metrics" | Self-Guided Reading Deck

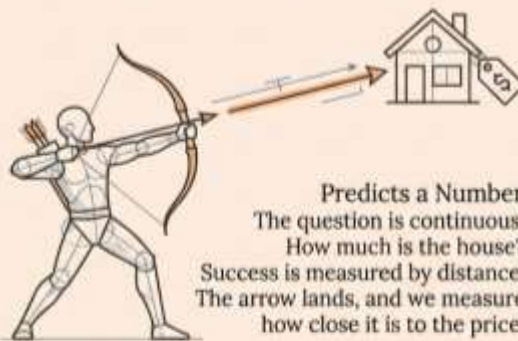
The Sentry vs. The Archer

The Sentry (Classification)



Predicts a Label
The question is binary:
Is it a wolf? Yes or No.
Success is measured by
accuracy and recall. The
bell rings, or it does not.

The Archer (Regression)



Predicts a Number
The question is continuous:
How much is the house?
Success is measured by distance.
The arrow lands, and we measure
how close it is to the price.

This deck focuses on The Archer. We are analyzing House Price Predictions, where the "Bullseye" is the actual sale price (represented by Earthy Brown #8B5E3C) and the "Arrow" is the model's prediction (represented by Pastel Orange #FFB380).

ML Model Evaluation Metrics

The Target Range: Five Houses, Five Predictions

To evaluate our model, we track 5 distinct properties. This 'Ledger' will be the foundation for every metric calculation.

Property ID	Actual Price (Target)	Predicted Price (Arrow)	Observation
House A	\$300,000	\$310,000	Small Overestimation
House B	\$450,000	\$440,000	Small Underestimation
House C	\$200,000	\$200,000	Perfect Shot
House D	\$600,000	\$550,000	SIGNIFICANT MISS
House E	\$350,000	\$360,000	Small Overestimation

Note House D. The model underestimated by \$50k. This outlier is the critical test for our metrics.

ML Model Evaluation Metrics

Measuring the Distance (The Residual)

The 'Error' is simply the mathematical distance between the Arrow and the Bullseye.

$$\text{Error} = \text{Actual Value} - \text{Predicted Value}$$

House	Actual	Predicted	Calculation	Error (Residual)
A	\$300k	\$310k	300 - 310	-\$10k
B	\$450k	\$440k	450 - 440	+\$10k
C	\$200k	\$200k	200 - 200	\$0
D	\$600k	\$550k	600 - 550	+\$50k
E	\$350k	\$360k	350 - 360	-\$10k

Problem: If we simply sum these errors (-10 + 10 + 0 + 50 - 10), the negatives cancel the positives. We need a way to summarize the magnitude of the error, not just the direction.

Mean Absolute Error (MAE): The Pragmatist

The Logic

Definition: The average of the absolute errors.

Characteristics: Treats all errors equally. It is robust to outliers and gives a realistic view of the “average” miss.

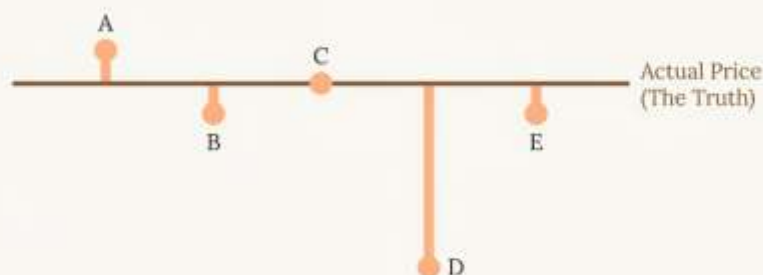
$$\text{MAE} = \frac{(\text{Sum of } |\text{Absolute Errors}|)}{\text{Number of Observations}}$$

The Ledger

House A:	$ -10 $	→	10
House B:	$ +10 $	→	10
House C:	$ 0 $	→	0
House D:	$ +50 $	→	50
House E:	$ -10 $	→	10
Sum:			80
Average (80 ÷ 5):			16

MAE = \$16,000. On average, our Archer misses the price by \$16k.

Visualizing the Absolute Error



MAE measures the average length of these orange bars. It ignores whether the bar goes up or down—it only cares about the length.

Mean Squared Error (MSE): The Punisher

The Logic

Definition: The average of the squared errors.

Characteristics: Penalizes large errors heavily. By squaring the error, a big miss becomes a massive number. Useful for optimization algorithms.

$$\text{MSE} = \frac{(\text{Sum of Errors}^2)}{\text{Number of Observations}}$$

The Ledger

House A:	$10^2 \rightarrow$	100
House B:	$10^2 \rightarrow$	100
House C:	$0^2 \rightarrow$	0
House D:	$50^2 \rightarrow$	2,500
House E:	$10^2 \rightarrow$	100
Sum:		2,800
Average (2800 ÷ 5):		560

MSE = 560. Note: The unit is now "Dollars Squared" (\$²). Hard to interpret, but mathematically powerful.

Root Mean Squared Error (RMSE): The Translator

We take the Square Root to bring the units back to reality.



The Comparison

MAE: \$16k The average absolute miss.	RMSE: \$23.66k The error penalized for the large outlier.
--	--

Why is RMSE higher? Because House D's large \$50k miss was squared, dragging the average up. RMSE is screaming that we have an outlier problem.

Choosing Your Scorecard: MAE vs. RMSE

MAE (The Steady Hand)



Behavior: Robust to outliers.

Use Case: Pricing mass-market condos. Being off by \$50k on one strange luxury house shouldn't panic the whole model. You want the general median accuracy.

RMSE (The Perfectionist)



Behavior: Sensitive to outliers.

Use Case: Pricing high-risk assets. A single massive miss could be catastrophic (like bank fraud or structural failure). You need a metric that reacts strongly to large errors.

R² Score: Beating the Baseline

Is the Archer better than a random guess?



R² (Coefficient of Determination) measures "Goodness of Fit". It doesn't tell you the **error in dollars**. It tells you how much better your model is compared to a naive baseline.

ML Model Evaluation Metrics

To compute the R^2 (Coefficient of Determination), we need to compare our model's error to the error we would get if we simply guessed the average.

- Think of R^2 as a score that tells us: "How much better is my model than a horizontal line representing the average?".

Steps:

1. Find the Actual Mean ($\bar{y}$)
2. Find **Unexplained Variation: Sum of Squared Residuals** (SS_{res}): This is the sum of the squared differences between the Actual y and Predicted $\hat{y}$
3. Find **Total Variation: Total Sum of Squares** (SS_{tot}): This is the sum of the squared differences between the Actual y and the Mean (the "Baseline" error).
4. The Final Result, we apply the R^2 formula:

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

ML Model Evaluation Metrics

Understanding the R^2 Ratio

$$R^2 = 1 - \frac{\text{Unexplained Variation}}{\text{Total Variation}}$$

Total Variation = How much house prices naturally jump around.

Unexplained Variation = How much error our Archer still makes (MSE).

Logic Flow

If Unexplained Variation is 0 (Perfect Model) $\rightarrow R^2 = 1 - 0 = 1.0$

If Unexplained Variation equals Total Variation (Useless Model) $\rightarrow R^2 = 1 - 1 = 0.0$

For our 5 houses, an R^2 of 0.75 means our model explains 75% of the price differences, leaving 25% to randomness or missing data.

ML Model Evaluation Metrics

Let's compute R^2 (Coefficient of Determination), for our 5-houses model.

1. Find the Mean ($\bar{y}$) First, we find the average of the actual house prices:

$$\text{Mean} = \frac{300k + 450k + 200k + 600k + 350k}{5} = \$380,000$$

To get a clear view of how our model is performing, let's consolidate our "Ledger" into a single master table. This allows us to compare our model's specific errors (SS_{res}) against the natural variation in the house prices (SS_{tot}) side-by-side.

ML Model Evaluation Metrics

Let's compute R^2 (Coefficient of Determination), for our 5-houses model.

the mean actual price ($\bar{y}$) is \$380,000.

Property	Actual (y)	Predicted ($\hat{y}$)	Residual ($y - \hat{y}$)	Squared Residual ($(y - \hat{y})^2$)	Mean Diff ($y - \bar{y}$)	Squared Total Error ($(y - \bar{y})^2$)
House A	\$300k	\$310k	-10k	100	-80k	6400
House B	\$450k	\$440k	+10k	100	+70k	4900
House C	\$200k	\$200k	0	0	-180k	32400
House D	\$600k	\$550k	+50k	2500	+220k	48400
House E	\$350k	\$360k	-10k	100	-30k	900
TOTAL				2800 (SS_{res})		93000 (SS_{tot})

ML Model Evaluation Metrics

Let's compute R^2 (Coefficient of Determination), for our 5-houses model.

Now we apply the R^2 formula:

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

$$R^2 = 1 - \frac{2,800}{93,000}$$

$$R^2 \approx 0.97$$

This means our model **explains 97%** of the variance in house prices.
In the world of real estate prediction, that is an incredibly high-performing model!.

ML Model Evaluation Metrics

Let's compute R^2 (Coefficient of Determination), for our 5-houses model.

Our model **explains 97%** of the variance in house prices.
In the world of real estate prediction, that is an incredibly high-performing model!.

Key Observations

- **Step 2 (SS_{res}):** This represents the "leftover" error our model couldn't explain. Notice how **House D** contributes the vast majority of our error (2,500 out of 2,800) because the mistake was squared.
- **Step 3 (SS_{tot}):** This represents how much the data "spreads out" from the average. If our SS_{res} was as high as 93,000M, our R^2 would be 0, meaning our model is no better than just guessing the average every time.

By comparing these two totals, we see that our model's error (2,800M) is tiny compared to the total variation (93,000M). This is why our R^2 score is so high (> 0.96)!

ML Model Evaluation Metrics

The Archer's Reference Guide

Metric	Role	Key Trait	When to Use
MAE	The Pragmatist	Robust to outliers.	For general accuracy interpretation.
MSE	The Optimizer	Penalizes large errors heavily.	For mathematical optimization (training).
RMSE	The Translator	Interpretable units (\$) but sensitive.	When large errors are costly.
R^2	The Benchmark	Measures fit (0 to 1).	To compare against a baseline guess.

ML Model Evaluation Metrics

Try
Yourself!

The Professor's Challenge: Applied

Imagine two different Real Estate Clients.

Client A: The Flipper



Context: Buys hundreds of cheap homes. Needs the portfolio average to be correct. One bad prediction doesn't hurt much.

Verdict: Use **MAE**. Focus on the median performance.

Client B: The Luxury Broker

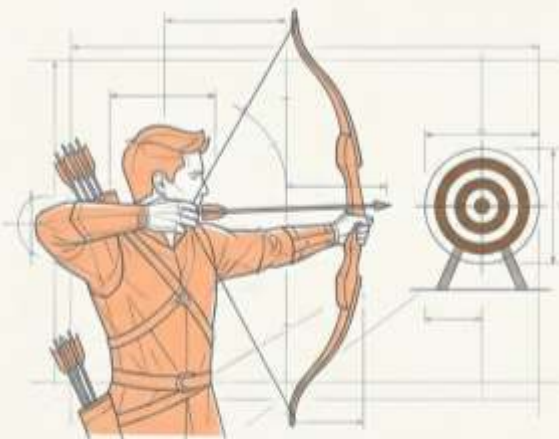


Context: Buys one \$10M mansion a year. A single bad prediction of \$1M is catastrophic.

Verdict: Use **RMSE**. Heavily penalize and detect any large deviations.

The "best" metric depends on the business cost of being wrong.

The Feedback Loop



The Archer does not improve by simply shooting arrows. The Archer improves by studying the ledger.

- To fix big mistakes, look at RMSE.
- To improve consistency, look at MAE.
- To prove value over guessing, look at R^2 .

Metrics are not the goal. They are the tools for improvement.

Overfitting and Underfitting

- In the world of machine learning, we often talk about how well a model **"generalizes."**
 - This means how well it performs on new, unseen data compared to the data it was trained on.
- Let's look at the two extremes **where generalization fails.**

Imagine you are studying for a math exam 📄🔍

Underfitting 📉 is like only skimming the chapter titles.

- You don't learn the actual formulas or patterns, so you perform poorly on the homework and the exam.

Overfitting 📈 is like memorizing every single practice problem exactly as it is written.

- You get 100% on the homework, but if the exam changes even one number, you fail because you didn't learn the underlying concept.

Overfitting and Underfitting

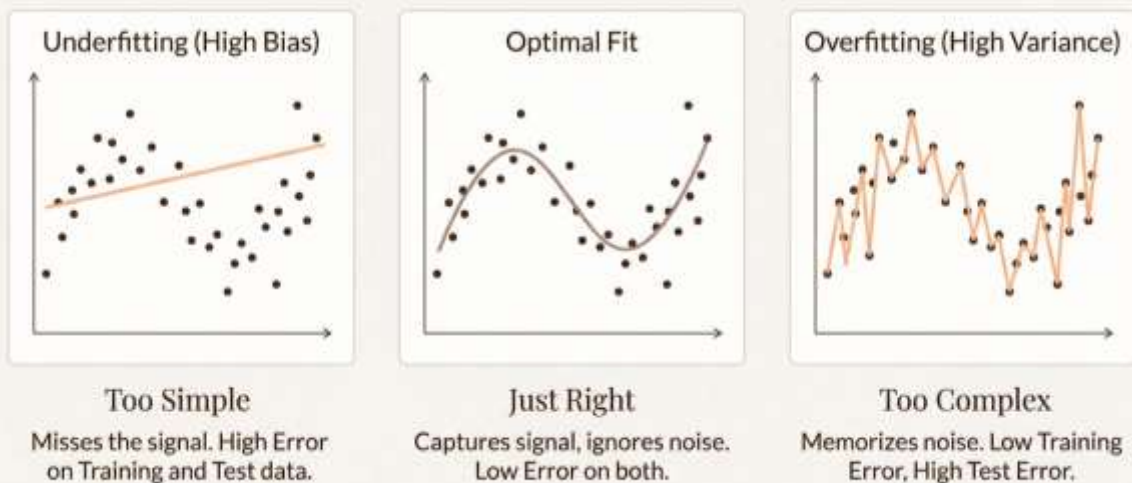
Concept	Definition
Underfitting	Occurs when a model is too simple to capture the underlying structure of the data. It performs poorly on both the training data and new data.
Overfitting	Occurs when a model is too complex and learns the "noise" or random fluctuations in the training data as if they were important patterns. It performs exceptionally well on training data but poorly on new data.

• The Bias-Variance Trade-off

- These concepts are closely tied to two types of error:
 - Bias:** Error from overly simplistic assumptions (leads to underfitting).
 - Variance:** Error from high sensitivity to small fluctuations in the training set (leads to overfitting).
- If you noticed your model had 99% accuracy on your training data but only 60% accuracy on your test data, which of these two problems do you think you are facing?*

Overfitting and Underfitting

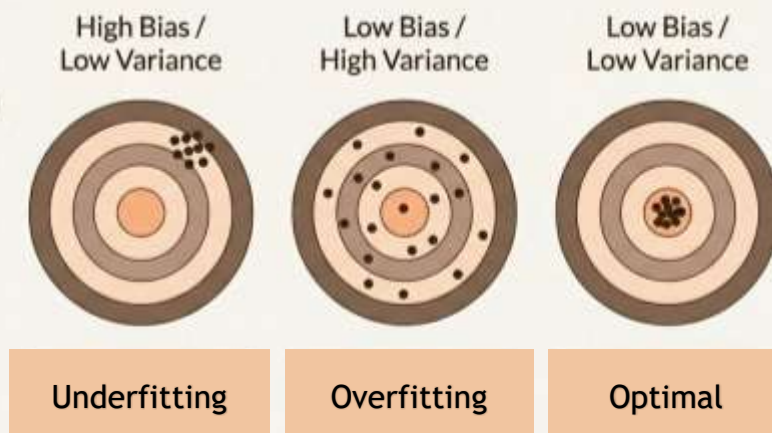
The Trap: Overfitting and Underfitting



Overfitting and Underfitting

The Bias-Variance Trade-off

- **Bias (Error from Assumptions):** Oversimplification. High Bias → Underfitting.
- **Variance (Error from Sensitivity):** Sensitivity to fluctuations. High Variance → Overfitting.
- **The Goal:** Minimize Total Error by finding the balance.



Overfitting and Underfitting

Target Mapping	Statistical Concept	ML State	The "Student" Analogy
 High Bias / Low Variance (Target 1)	Underfitting	The model makes strong, wrong assumptions. It is consistently off-target in the same way.	A student who only studied one chapter. They get every question about that chapter right but miss everything else consistently.
 Low Bias / High Variance (Target 2)	Overfitting	The model has no "fixed" wrong assumption, but it is extremely sensitive to noise. The arrows are all over the place.	A student who memorized the exact practice problems. If the exam question is slightly different, their answer "jitters" wildly.
 Low Bias / Low Variance (Target 3)	Optimal	This is the "Goldilocks" zone. The model has learned the true pattern and generalizes well to new data.	The student who actually understood the underlying math. No matter how you phrase the question, they hit the mark.

Overfitting and Underfitting

Why "High Bias" is Underfitting:

In Target 1, the model is "**biased**" toward a specific (wrong) area. It lacks the complexity to even reach the bullseye. This often happens with simple **Linear Models** trying to fit complex, curvy data.



Why "High Variance" is Overfitting:

In Target 2, the model changes its mind too easily based on small fluctuations in training data (high variance). It "**over-learns**" the noise, leading to the scattered pattern you see on the target. This is common in **Non-parametric models** like deep decision trees.



The Goal (The Bullseye):

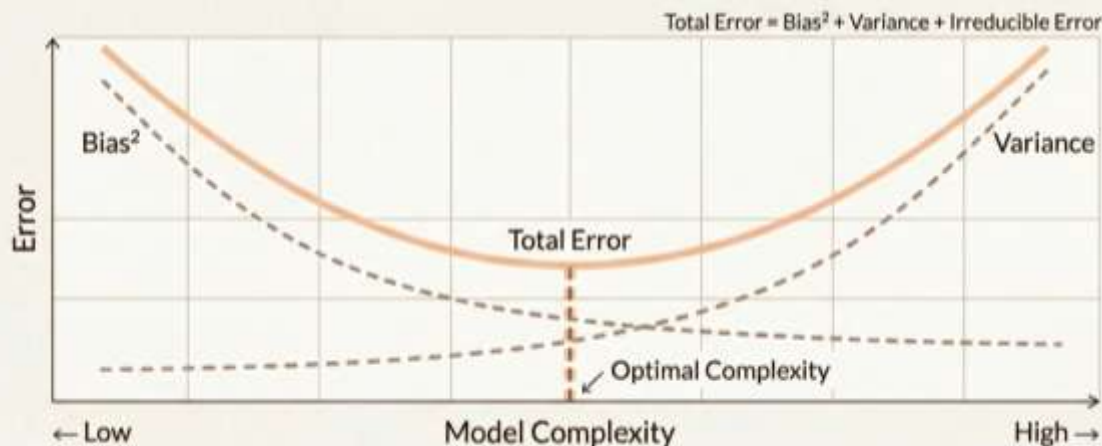
Our job is to find the **sweet spot** where we minimize both. We use **Evaluation Metrics** (like the R^2 we calculated) to see which target our model currently looks like.



Machine Learning Workflow

Decomposing Total Error

Finding the Sweet Spot of Model Complexity



Real-World Applications & Use Cases

Healthcare



- Disease diagnosis (Classification)
- Tumor detection in X-rays (Image Analysis)

Finance



- Credit scoring (Regression)
- Fraud detection (Anomaly Detection)

E-Commerce



- Recommender Systems ("Customers who bought X...")
- Customer Churn Prediction

NLP



- Spam Filtering (Classification)
- Sentiment Analysis & Chatbots

Summary: From Data to Generalization

- 1. Shift to ML:** We utilize data to find patterns, rather than hard-coding rules.
- 2. Model Choice:** Depends on structure (Parametric vs. Non-Parametric) and goal (Generative vs. Discriminative).
- 3. Validation:** The Train-Test split is non-negotiable for testing real-world performance.
- 4. The Balance:** Success lies in managing the Bias-Variance Trade-off to avoid Overfitting and Underfitting.

Closing...

Course Outcomes & Learning Path

What we accomplished today and what comes next



CO1: Foundations

Defined ML, AI, and DS. Explored the history and taxonomy of learning.



CO2 & CO3: Data Skills

Next Steps: Mastering Pandas and performing deep EDA on datasets.



CO4 & CO5: Math & Stats

Future Units: Linear Algebra, Probability, and Optimization techniques.



CO6: Model Understanding

Goal: Learning how to choose the right algorithm for the data.

Closing...



In the Next Class:

Unit-1, Part-2: NumPy for Numerical Computing (Practical)

Please bring your PCs in class